

FOOD REMEDY API

RE011 – Understanding Hormonal Fluctuations and Their Biological Role

Prepared For: Food Remedy API – Research Team

Author: Sehajpreet Singh

Date: March 2026 (T1 2026)

1. Introduction: -

Hormones are like chemical messengers which are very essential and get produced by the endocrine glands. They play a very important role for the psychological processes like the metabolism and energy balance. Having the understanding about the hormones fluctuation and how the hormones interact themselves with the nutrition is very essential to have a basic understanding and developing the personalized dietary systems. Not only they serve but having the understanding about the hormonal fluctuations will help the food remedy API Project to make a better dietary system for different age groups by undertaking all the essential requirements.

2. Key hormones involved in metabolism and nutrition: -

2.1 Insulin

- The production of the insulin gets from the pancreas
- The main function of the insulin is to regulate the blood glucose levels so that the cells become able to absorb the glucose properly
- It also promotes the storage of the glycogen as well as the other storages for the fat
- The carbohydrate levels in insulin are high as the intake of insulin increases

2.2 Cortisol

- Cortisol is also known as the stress hormone and the cortisol is released by the adrenal glands
- The main function of the cortisol is to promote the storage of fat
- If the stress levels are high then the cortisol also increases the level of glucose in the blood
- If the cortisol is high then it will increase the cravings for sugary food and to balance the level of cortisol it is very important to intake protein rich food

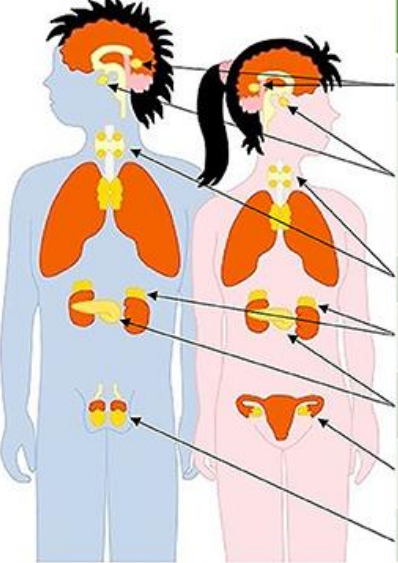
2.3 Estrogen

- The primary sex hormone for the female is estrogen
- Also it provides the influences for the fat distribution, sensitivity for the insulin as well as the appetite regulations
- If the estrogen in decline level then it automatically increased the storage of fat

- To maintain the estrogen it is very important for the individual to take a balanced diet for the hormonal stability.

2.4 Thyroid hormones: -

- The main purpose of the thyroid hormones is to have a control over the metabolic rate which is also known as basal metabolic rate
- Main function of these hormones is to regulate and analyze that how quickly the body is using the energy
- There are a few essential things that are very essential for the functioning of thyroid like the Iodine and zinc
- If the nutrition intake is not good then it will automatically slow the process of metabolism.



Gland	Hormone	Target Organ	Function
Pineal gland	Melatonin	Many	Biological clock
Pituitary gland	FSH / LH ADH Growth hormone Oxytocin Prolactin	Ovaries Kidneys Uterus Breast tissue Many others	Menstrual cycle Osmoregulation Growth & division Birth contractions Milk production
Thyroid gland	Thyroxin	Liver	Metabolic rate
Adrenal glands	Adrenaline Cortisol	Many	Fight or flight Anti-stress
Pancreas	Insulin Glucagon	Liver	Blood sugar levels
Ovaries	Estrogen Progesterone	Uterus	Menstrual cycle
Testes	Testosterone	Many	Male characteristics

3. Hormonal fluctuations across different conditions: -

3.1 Daily cycles also known as the Circadian Rhythm

- The peak time of cortisol is in the morning which is very essential to boost up the energy
- The sensitivity level of insulin is very high during the daytime hours
- Also in the cycle it is very important role of melatonin which rise in the night and helps to get the better sleep
- The main impact on the nutrition to eat early in the morning so that it will help to improve the process of metabolism
- If the individual is eating during late nights then it led to increase in the storage of fat

3.2 Stages of the life

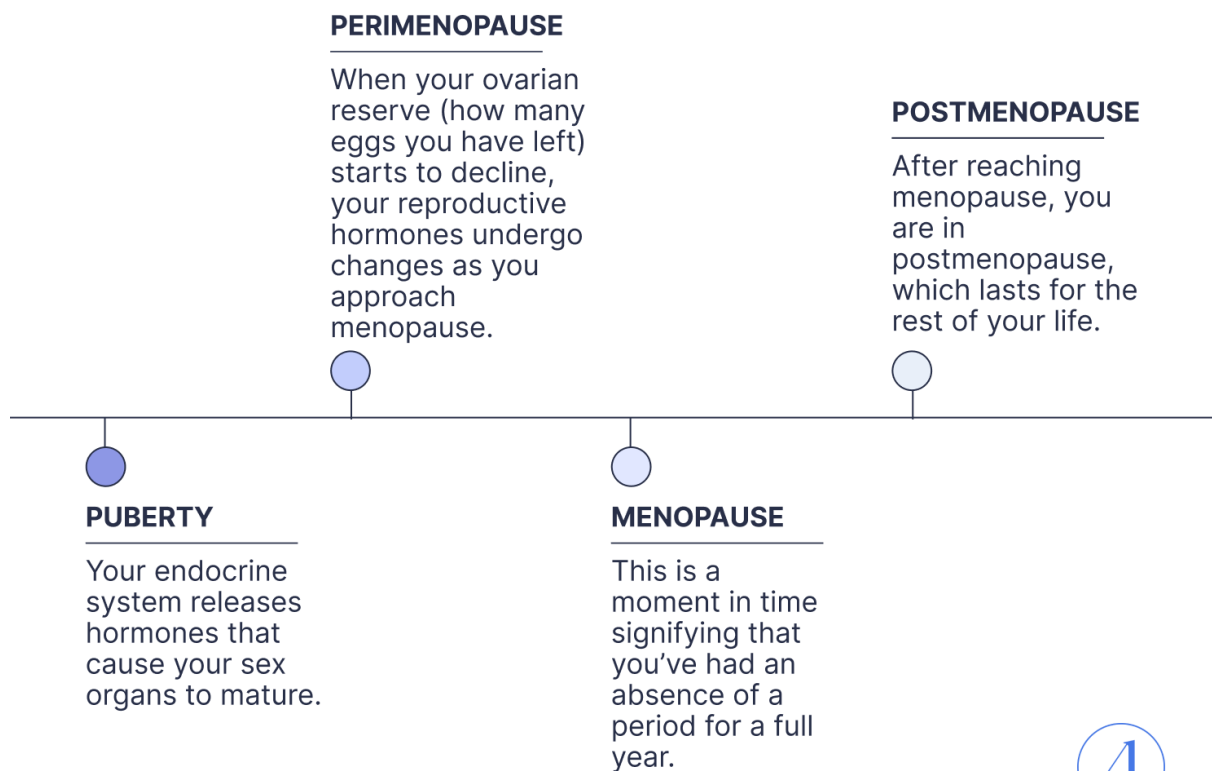
- In the stage of **adolescence** the hormonal Surges always led to increase in the level of appetite as well as the needs of energy.
- During the **Adulthood** there is a as well as the stabilization in the hormonal fluctuations and the overall lifestyle of the individual is influenced by the metabolism.
- In the **old age group** the estrogen levels always decrease and also metabolic rate gets reduce.

3.3 Stress conditions

- If there are stress conditions like the chronic stressor then it can easily lead to the sustainability of cortisol elevation
- Also it increase the appetite, eating emotionally which will lead to the increase in the storage of the fats in the body.
- Any who is under the stress conditions always feel the craving to eat high sugar content food as well as the high fat content food.

Hormonal fluctuations throughout your life

Hormone levels, especially those tied to reproduction, fluctuate throughout the lifespan, as well.



4. What is the working of hormones and energy balance, appetite and metabolism: -

4.1 Regulations of the appetite

- In the appetite regulation the cortisol always impacts increase the level of hunger
- The estrogen led to the suppresses appetite
- The impact of the insulin is to give the signals related to satiety for the short period of time

4.2 Energy balance

- The energy balance mainly controlled by the hormones that are regulated by the intakes as compare to the expenditure.
- If there are stress conditions than the cortisol releases the energy stored.
- The thyroid hormones leads to increase in the energy expenditure
- Insulin helps to store the energy which includes the fat as well as the glycogen.

4.3 Metabolism

- Thyroid hormones helps to have the control over the metabolic speed
- Insulin helps to determines the nutrient storage rather than determining the usage
- Cortisol affects the distribution of the fat content in the body

5. Health effects of hormonal imbalances: -

5.1 Thyroid disorders

- Hypothyroidism: -
 - Slowing the process of metabolism
 - Leading to the symptoms of fatigue
 - Also lead to the weight loss
 - Rapid metabolism

5.2 Estrogen imbalance

- It led to the mood swings
- Also lead towards the gain in the weight
- Increase in the risk related to the metabolic issues

5.3 Insulin imbalance

- Imbalance of the insulin in the body do have impacts related to type two diabetes as well as the obesity
- The imbalance of insulin in the body is caused by the excessive intake of sugar content into the body.

5.4 Cortisol imbalance

- Cortisol imbalance is also caused due to the chronic stress
- It led to the gain in the weight of the individual person
- It impacts the sleep cycle of that particular person
- Also the Anxiety levels goes high due to imbalance of cortisol.

Hormone	Function	Effect on Nutrition
Insulin	Glucose regulation in the body	Fat storage and sugar control

Cortisol	Stress response and chronic stress	Increases cravings for the sugary food
Estrogen	Fat distribution in the body	Appetite regulation
Thyroid	Controlling the Metabolism	Calorie burning

6. Conclusion:

To conclude I want to mention that after doing the research on hormonal fluctuations I become able to have the understanding that how the hormones play very important role in the regulation of metabolism, appetite as well as the balancing of energy. Also the fluctuations in the daily sleep cycles and the stress conditions get influenced by the nutritional needs. Having the proper understanding about this concept is very essential for the Food Remedy API project to do the improvements in the personalized dietary system for the public.

7. References: -

- World Health Organization 2023, Healthy diet, World Health Organization, viewed 10 April 2026, <https://www.who.int/news-room/fact-sheets/detail/healthy-diet>
- National Institutes of Health 2022, Endocrine system and hormones, National Institutes of Health, viewed 10 April 2026, <https://www.niddk.nih.gov/health-information/endocrine-diseases>
- Harvard T.H. Chan School of Public Health 2023, The nutrition source: hormones and weight, Harvard University, viewed 10 April 2026, <https://www.hsph.harvard.edu/nutritionsource/>
- Better Health Channel 2024, Hormones and the endocrine system, State Government of Victoria, viewed 10 April 2026, <https://www.betterhealth.vic.gov.au>
- National Health Service 2023, Hormonal health and metabolism, NHS, viewed 10 April 2026, <https://www.nhs.uk>